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1.- OBJECT AND SCOPE

This Job Instruction has the purpose of establishing the technical basis of design, layout and installation of all piping systems in process units, utility services and storage farms.

This Job Instruction is part of the Quality System of **Intecsa Industrial** and has been prepared considering the ISO 9001.

2.- STANDARDS, REGULATIONS AND APPLICABLE CODES

References to any standard, regulation or code always correspond to the latest revision.

Unless notes otherwise, the design, materials, fabrication and inspection shall comply with the requirements of the applicable code, according to the type of project listed below:

- ASME B31.3 Process Piping
- ASME B31.1 Power Piping
- ASME B31.8 Gas Transmission & Distribution Piping System
- ASME B31.12 Hydrogen Piping & Pipelines

The dimensions, materials, tolerances and marking of metallic pipe components will agree, among other things, with these standards or codes:

- American Petroleum Institute - API.
- American National Standards Institute - ANSI.
- Manufacturer´s Standardization Society - MSS.
- American Society for Testing and Materials - ASTM.
- American Society of Mechanical Engineers - ASME.

Piping systems and their components that are not covered by a recognized international code or any other authorized standards, will follow engineering standards and generally accepted industrial practices.

All requirements of the “Reglamento de Instalaciones Petrolíferas” or the equivalent of the country where the facility is located, if applicable, must be complied with.

3.- PIPING MATERIAL SPECIFICATIONS

Material specifications are the set of pipe material classes.

Piping material classes shall be calculated for maximum design pressure and temperature conditions, corrosion allowance, and compatibility with the fluids in question.

The design pressure and temperature used for the selection of materials and standardized elements that compose the piping networks, will comply with the requirements of ASME B 31.3 standard, the European Parliament and Council Directive 97/23/CE (PED), complementary instructions (ITC's) and the “Pressure Equipment Regulation - PER”, (provided that the site where the facility is to be built belongs to the European Economic Community).

The design pressure of each component in a piping system, shall not be less than the pressure at the most severe condition coincident with the internal or external pressure and temperature (minimum or maximum), expected during operation.

The design temperature of each component in a piping system, is the temperature at which, under a coincident pressure, the highest thickness or rating of component is required.

The pressure rating of each piping class specification will be selected based on the maximum design pressure and temperature combination, in accordance with ASME B16.5. "Steel Pipe Flanges and Flanged Fittings".

The pipe thickness indicated in each class of materials shall be the thickness required for the design conditions. This thickness shall be calculated considering the corrosion allowance and the manufacturing tolerance according to the standard applicable to the project (ASME B31.1, ASME B31.3, ASME B31.8, ASME B31.12).

The pipe material will be selected in accordance with ASTM "American Society for Testing and Materials" and API "American Petroleum Institute".

All threads on pipe ends and flange and fitting connections shall be NPT, in accordance with ASME B1.20.1.

Forged fittings with threaded and socket type (SW) ends shall be in accordance with ASME B16.11.

Beveled butt weld (BW) ends shall be in accordance with ASME B16.25.

In general, all fittings (elbows, tees, reducers, etc.), shall follow the requirements established for pipe regarding strength, thickness and type of ends.

Prefabricated materials and subcontracted packages in accordance with vendor standards will be subject to review and acceptance of **Intecsa-industrial**.

The piping class specifications for each project, will in any case govern, the criteria for the selection and use of the elements to be used in piping systems.

4.- PLOT PLAN

- General arrangement and distances between facilities

  The general layout criteria, as well as the minimum distances between the various installations, and between these and the outside, will comply with the requirements and recommendations of the Client, of the "Regulation of Petroleum Installations", or the equivalent of the country where the installation is located, if applicable. Always considering the prevailing wind direction of the site.

- Access

  The entire facility will be surrounded by an enclosure with a minimum height of 2.5 meters. The access from external roads will be located and designed appropriately to facilitate any intervention and evacuation, if necessary, in addition to allowing vehicle entrances and exits free of maneuvers.

- Roads

Inside the facilities, roads will have the following width:

a. Roads of free circulation: 6 meters

b. Restricted or regulated circulation roads: 4 meters

All units that integrate all the facilities will have service roads around them, at least 4 meters wide, counted from the unit boundaries.

In any case, the turning radius of all roads will allow the easy circulation of vehicles.

- Distance between equipments

In general, the minimum distance between the equipment comprising the facility will allow the necessary requirements for maintenance and / or the free circulation of the operating personnel, including, if necessary, the mobile elements or vehicles used.

As particular cases, the following are listed:

a. Combustion equipment (furnaces, boilers, etc.)

The minimum distance between combustion equipment and other equipment of the facility that operates with hydrocarbons (towers, vessels, pumps, compressors, air-coolers, exchangers, etc.) shall be 15 meters, measured horizontally.

However, the preheat exchangers and reactors will be located as close to the furnace to allow their maintenance and operation. A distance of at least 5 meters is recommended.

b. Air coolers

The air coolers may be placed on the pipe-rack, but in this case, if they operate with hydrocarbons, no equipment, normally pumps shall be placed under them. Except, if the floor of the air cooler support elevation is made of blind tramex (blind floor).

c. Storage tanks (cylindrical and spherical tanks)

Generally, storage tanks are arranged in parks, trying to gather those containing products of the same class, within the same dyke.

The arrangement and spacing of storage tanks will comply with the requirements and recommendations of the Client, of the "Regulation of Petroleum Installations" or the equivalent of the country where the installation is located, if applicable.

For more information on equipment's layout see IT-TU-16 of **Intecsa-industrial**.

5.- GENERAL PIPING DESIGN

- The piping sizes 1 1/4", 2 1/2", 3 1/2", 5", 7" and 9" will not be used except for connections to equipment or special services.

- Typical pipe diameters are ½”, ¾”, 1”, 1 ½”, 2“, 3”, 4”, 6”, 8”, 10”, 12”, 14”, 16”, 18”, 20”, 24”, 26”, 30“, 32”, 36”, 40”, 42“, 44”, 48“, 52”, 54“, 56”, 60”.

- On piperacks / pipeways the smallest pipe diameter should be 2”.

- Flat face flanges will be installed against valves and equipments with flat faces.

- For design of underground networks see IT-OC-80 of **Intecsa- industrial**.

6.- DESIGN IN PROCESS AREAS

- The connection between units, equipment location and piping layout will be designed based on economics, safety, ease of maintenance and operation. Equipment and piping layout shall be designed to provide an orderly appearance. In general, it is preferable that all main process and service lines are carried on piperacks. Approximately 20% of the width of the piperack is left free for future lines. In case where process lines need to be routed below ground, they will be carried in visitable trenches, underground process lines are not permitted, except for drainage circuits. Cooling water pipes can be above or below ground depending on the economic study or other needs.

- No additional space shall be provided for future equipments, units, etc., unless specifically required by the customer or process department. This requirement must be indicated on the plot plan and on the P&ID’s.

7.- DESIGN OUT OF PROCESS AREAS

- The pipes should normally be installed on trays, resting on sleepers. The elevations of the sleepers will be staggered to allow an easy crossing or a change of direction. Flat turns may be used when whole sections of trays change their direction on sleepers.

- The height of the sleepers will be enough to maintain a minimum spacing between the bottom of the pipe and the floor of 300 mm.

- When a pipe track needs to cross a road, it will be made in such a way that a free height of 6 m is left above the grade, the height will be determined by the clearance required by the Customer. If this is not possible, it will pass under the road though a culvert or by a conduit pass pipe, with a minimum depth of 600 mm to the upper generatrix of the pipes.

- When it is necessary to have another level, a steel structure will be used on the sleepers, with 900 mm minimum between the two levels. Considering the largest pipe diameter and insulation thickness to fix the final height.

- Attention will be paid to the possible thermal expansion of pipes, because of weather conditions or process conditions. The expansion of the pipe will be preferentially absorbed using expansion loop. Avoid as much as possible the use of expansion joints.

- Tank's piping will be provided with the necessary flexibility to absorb the settling of the tanks.

8.- ACCESS, HEIGHTS AND SAFETY CLEARANCE

- Heights and safety clearance and access to equipments, structures and supports shall be in accordance with Tables of the IT-TU-16, RULES FOR LAYOUT AND EQUIPMENT ARRANGEMENT DRAWING, of **Intecsa-industrial**.

- Unobstructed access way from buildings, overhead equipment or structures shall be provided as exit routes.

- Two access ways (i.e., two stairs or a stair and a monkey ladder) shall be provided for any elevated platform serving three or more vessels and for the battery limits unit platform.

- Secondary emergency exit shall also be provided when the distance to be traveled from the farthest point of a platform to an exit is more than 15 m.

- In towers and vessel platforms monkey ladders will be installed, using stairs for steel structures platforms. Monkey ladders and stairs should appear in the plot plan.

- Monkey ladders will have safety cages and closures as required in the standards of **Intecsa-Industrial.**

- In case of having block valves at the battery limit unit, a platform with stairs shall be provided for their access and operation. The battery limit unit shall be at least 15 m., measured horizontally, from any equipment in the unit.

9.- MAINTENANCE

- Equipments, structures and piping shall be arranged in such a way that they can be maintained and serviced through mobile devices. Fixed ways for maintenance, when explicitly required, will be indicated on the plot plan.

- Vessels with removable internal elements or charging/discharging nozzles will have a reserved area for operation and access.

- Exchangers with removable tube bundles shall have a free area of equal length to the tube bundle, sufficient for disassembly.

- In furnaces, sufficient free areas shall be provided for the extraction of pipes, soot blowers, filters, burners, fans and other accessories related to this equipment.

- During maintenance operations, spectacle blind will be used to isolate pipes according to ASME B16.48.

- For the maintenance of spectacle blind, permanent access is not required but must be accessible by temporary scaffold or portable ladder.

10.- EQUIPMENTS

10.1. Pumps

- The suction pipe of the pump should be as short and direct as possible and never smaller in diameter than the pump nozzle. If a reducer is required at the suction connection, it shall be eccentric type and shall be placed with the flat side up.

- Pump piping should be sufficiently flexible and adequately supported to avoid stresses and strains in their nozzles that could damage the pump casing.

- In the connection to the pump, it will be installed of the eccentric type, flat face up.
  In all cases, space shall be provided near the inlet nozzle for the installation of a temporary filter.

- Piping should be designed and constructed to allow disassembly of the engine. Pumps with side suction, disassembly of the pump casing should also be permitted, while the suction and discharge valves remain in place.

- The suction valve is normally a gate valve with the same size as the nozzle of the pump. On the discharge side, a check valve should be installed between the pump and the shut-off valve on centrifugal and rotary pumps. In positive-displacement pumps, a safety relief valve should be installed between the pump and the shut-off valve.

- Pressure gauges shall be installed on all pump discharges between the nozzle and the first valve.

- Pump auxiliary piping, for sealing and/or cooling, shall be designed so as not to interfere with the pump, complying with the systems and schemes specified in API 610.

10.2. Exchangers

- Heads and shells shall be free of piping, structural components, etc., for bolts removal from the exchanger flanges and extraction of the heads and the tube bundle.

- Avoid the assembly of valves directly to the nozzles of the head, if they obstruct the removal of the same.

- Exchangers can be located as single items, in pairs (this is the most common installation), or in larger groups when no intermediate control is required between the shell and tube streams. Paired exchangers may operate in series, parallel, or dissimilar services.
- Horizontal shell and tube exchangers should be positioned so that the channel end faces the auxiliary road or maintenance access way for tube bundle removal with adequate space provided at the front end of the exchanger for bonnet removal.

- In the orientation of the pressure and temperature taps, which are usually placed on the nozzle of the exchanger, it is necessary to consider the piping layout, and that the temperature tap should have a clear area for thermocouple dismantling.

- Avoid placing pipes on the longitudinal axis of an exchanger when it has a monorail above it, to allow its free movement.

- As a rule, the fixed support will be the closest to the pipes that are buried, but if they all come from a pipe tray, it will be the one closest to it. However, this will be determined by the flexibility analysis specialist.

- In the case of a reboiler located at the bottom of a column, the elevations indicated on P&ID will be studied to fix the reboiler supports. If it is equipped with levels, they will be read from the position of the control valve.

- In cold weather, it is convenient to install a bypass between the inlet and outlet water lines, before the block valves, to allow water flows outside the exchanger in case it is out of service, avoiding the freezing of the water in the pipes.

10.3. Aircoolers

- If the equipment has several inlets and outlets, the piping arrangement should be designed symmetrically so that the flow is uniform. The flow distribution shall be according to P&ID.

- In general, no piping and/or metallic structures should pass over the equipment, for easy removal of the tube bundle of it.

- Manufacturer's recommendations will be considered when designing the maintenance and operation platforms (engines and heads).

10.4. Vessels

- Platforms for vessels will be the minimum necessary, they will cover the operation and maintenance requirements for manhole, valves, instruments, etc... However, it is permitted to operate standpipe valves from ladders.

- In case of vertical vessels with flanges on top, the platform should cover all flanges for removal.

- In case of multiple vessels together, we must ensure that the manholes are in the same position and on the access side.

- The inlet and outlet connections are positioned at opposite sides; the inlet at the top and the outlet at the bottom. The level indicators will be located away from areas of turbulence and eddies produced by the inlet and outlet nozzles. The pressure connections are placed on top, and the temperature at the bottom near the outlet, or in the same line. These general rules will always apply if there are not specified otherwise for process reasons.

- To avoid structures in elevated horizontal vessels, it is possible to extend their support, supporting the necessary platform for maintenance and operation. In the case of multiple vessels, common platforms may be used, and in case of three or more, by placing two ladders, one at each end.

10.5. Towers / Column

- On towers, piping shall be located where possible, radially around the tower on the pipes tray side; manholes and platforms shall be located on the access side. Manholes will be on or around a common axis.

- Adequate space must be provided between piping and between the back of piping and the tower shell to facilitate the installation of pipe supports, which are attached to the tower. Also, should be considered for this distance the thickness of the insulation.

- Platforms shall be located at a height such that the distance between the platform and the manhole axis is 900 to 1500 mm.

- Monkey ladders may not have more than 9000 mm between platforms.

- Common platforms will be provided where so dictate practice, or the economy in adjacent towers with common manhole elevations.

- Davits will be provided for the maintenance of internal elements, safety valves, etc. in those vessels where it is necessary for height or inaccessibility of the mobile cranes. Davits are oriented to permit transportation of accessories within the access area.

- Instruments of level, pressure and temperature will be accessible from platforms or ladders.

- The orientation of the nozzles of the tower will be in accordance with the elements of the internal parts and the equipment located on its sides, contributing to a good arrangement of piping between this and that equipment.

- Safety valve for protection of the tower is usually connected to the steam line of the tower head. In the case of an open system, the safety valve must be placed on the highest platform of the tower. In a closed system with safety valve discharges, you should choose the lowest platform above the discharge collector.

10.6. Furnaces

- In furnaces with bottom-mounted burners, a clear headroom of at least 2400 mm. should be provided. The pipes feeding the burners are an exception for reasons of accessibility, but these will allow an easy escape in an emergency.

- Furnaces and other combustion equipment must be located upstream of the prevailing wind direction of equipment operating with hydrocarbons.

- The quench steam manifolds, and fuel gas shut-off valves shall be located at least 15 m from the furnaces they protect.

- In furnaces with bottom-mounted burners that are fueled by fuel oil and fuel gas, the fuel shut-off valves will be operated from the sight glass observation deck.

- The pressure relief doors and access doors of pipes must be free of obstructions. The pressure relief doors are oriented such that they cannot affect the adjacent equipment.

- The service subcollectors will be located above the burners and holes of pilot ignition and observation of the flame of burner. The connections to these burners will be made at the top, except fuel that will be made underneath. The connections of the burner will be flanged for easy disassembly.

- The fuel oil subcollectors will have connections that can be cleaned and have no dead ends, getting a "by-pass" which allows a continuous movement of fluid.

- The inlets and outlets of product in furnaces have a symmetrical layout, achieving a uniform flow in all steps.

10.7. Compressors

- Suction lines to compressors, centrifugal or reciprocating, will avoid any pocket which condensate could be accumulated and will have the fewest number of bends. The separator tank shall be located as close as possible to the compressor suction.

- In the case of reciprocating compressors space for the extraction of the pistons shall be provided and will follow the recommendations of the vendor for maintenance.

- The pipes should never be supported on the equipment, giving them the necessary flexibility, even using expansion joints to absorb vibration of the compressor.

- For air compressors, the intake, although outside the building should be as close to the equipment and prevent polluted or dusty atmospheres.

- When the compressor operates with gas heavier than air, ditches and low spots in the pavement should be excluded, which could store gas pockets that would cause gas poisoning or explosions, although adequate ventilation.

- Reciprocating compressors in line will have a partial platform to accessing to cylinder valves if required.

- Centrifugal compressors will have complete operationally platforms.

10.8. Steam Turbines

- The steam supply to the turbine, will take a trap station and steam trap and a filter (unless incorporated), located as close as possible to the nozzle of the turbine.

- It is desirable to have a "bypass" seat valve for heating the machine. The inlet valve is mounted horizontally.

- The exhaust pipe will take check and block valves close to the turbine to allow easy handling. Among these valves and the outlet nozzle is normally installed a safety valve. Also a steam station and a steam trap will be installed.

10.9. Storage tanks

- Generally storage tanks are arranged in bunds. Taking into account the product to be stored, several tanks could be cluster in one bund (provided that the fluid tanks are compatible). Both the size of the bund as the distance between tanks and between the storage set itself with other facilities, will be calculated according to the relevant regulations, (Regulation of chemical storage, regulation of refineries ...).

- Pipes are always designed on the ground and go grouped inside the bund resting on sleepers.

- When designing the pipes the temperature is taken into account, so that the installation of a loop into the bund is not necessary.

- Pipes that connect in the top of the tank, will be grouped together to support them jointly on the wall of it.

- The manholes will be located diametrically opposite each other (shell and roof), to facilitate ventilation in maintenance operations.

- The nozzles located on the roof shall be located in the access platform and shell nozzles are directed towards the pumps, designing lines in both cases, grouped in the same area of the tank.

- The valves are installed outside the bund and only when necessary in its own tank nozzle. These valves will be operated by remote control (as indicated on the P & ID).

- The platforms on the roof of the tank will cover only the part needed for access and operation of valves, instruments and nozzles.

- Helicoidal stairs shall be placed upstream of the prevailing wind direction.

10.10. Spherical Tanks

- For storage of “A” class liquefied products and “A1” and “A2” subclasses, which influence the pressure and temperature, spherical tanks are used (pressure spherical vessels).

- The spheres will be placed given the current regulations for the products listed above.

- Considering that the stored products are highly volatile, it is not necessary bunds to full capacity. Bunds with liquid flow capacity for the non-volatilized will be available, as indicated in the corresponding regulations.

- The inlet and outlet nozzles of product are placed only on the top and bottom respectively. Connections will not be allowed in the rest of the equipment.

- Pipes are designed on the perimeter of sphere and in the vertical plane, as close as possible to it so as to facilitate its anchorage, using for 45 degree elbows.

- It is acceptable as long as the diameter permits use curved pipes.

- Pipes are designed to the side of the rack or the track that connect with.

- Each sphere will have a single access ladder and will be oriented to the road or to right perimeter access, so as to allow rapid and safe evacuation of the area.

- Upper platform will cover the necessary area for the maintenance of the valves and instruments installed.

11.- PIPING LAYOUT

- All pipes will be designed attempting to get the shortest possible routes and the fewest number of accessories, respecting the limitations given by the thermal expansion and flexibility inherent in its design.

- All pipes situated inside units will be grouped and supported on supports, preferably high. They will be installed so as to allow easy and complete access to the equipment for operation and maintenance.

- Piping routes will be designed so as to eliminate liquid and gas pockets. Pockets must be avoided particularly in two-phase fluid and in lines of condensable steam.

- When a bundle of pipes or some of them, even when located at the end of the tray, change its direction, it will change in elevation. The expansion loops will also follow this rule.

- In pipe trays spaces must be provided for electrical and instrument trays if required. The separation between pipes is according to Standard SH-9 **Intecsa-Industrial**, leaving a 20% of free space for future.

- Vertical pipes in vessels are preferably installed near the wall thereof, to be supported on them.

- When the pipe passes through floors, roofs or walls, wall-through (bushings) are placed with enough size to allow movements caused for expansion, over insulation, etc.

- Minimum straight distance of pipes, upstream and downstream of an orifice plate free of turbulence will be according to the standard SH-29 of **Intecsa-Industrial**, which also indicates the position of the intakes depending on the type of fluid.

- All instruments with readings will be placed accessible from platforms or ladders; those who send signal only be accessible from a portable ladder.

- The pipes without isolation will be directly supported on the bracket, while pipes with insulation will be supported using steel shoes. The shoes will be high enough to save the insulation, as indicated in the standard of **Intecsa-Industrial**. The shoes will usually be welded to the pipe or with clamps.

- The service pipes must be located in the upper tray and the steam, condensate and cooling water collectors, preferably on the sides of the tray.

- Reserve spaces around pillars of the pipe tray and structures for assembly elements like rises of vents, service stations, electrical panels or boxes, control valves, samplers, steam traps, steam and condensate manifolds, etc.

- Good coordination will be basic to avoid frequent interference between specialties, in the situation of these elements.

- Designs made in 3D, all the listed items will be modeled in its actual location.

12.- FITTINGS

They will be always included in the piping material specifications of the project. However, as a general rule you can follow the following recommendations:

- Elbows will be with a radius 1.5 times its diameter, the pipe can be bend with a radius equal to 5 diameters. Galvanized pipe or internally coated is not allow bend. To use standard view miter elbows view the standard SH-1 of **Intecsa-Industrial.**

- The change of the pipe diameter will be made by reducers and swages, avoiding reducing flanges.

- All drills for bolts in the flanges go riding on the main axes. When this is not possible be shown in a detail isometric with its exact position of the drill holes.

- Flat face flanges will be used whenever the nozzle of the equipment flange is flat face, and its gasket will be the outer diameter of the flange (with drill holes for the bolts).

- Neck flanges "welding neck" is preferably used against the flat "slip-on".

- Blind disks are accessible from ground or platforms. Disks located in trays of pipes are considered accessible. Disks weighing more than 45 kg must be accessible with mobile devices. But if this is not possible, have davits. When multiple blind disks are grouped will be placed in a phased manner.

13.- VALVES

- Frequent use valves or requiring maintenance, shall be accessible from the ground, platforms or ladders. It is considered accessible a valve which stem is horizontal and is less than 2 m of the operating level.

- 2" and larger valves, which are not removable from the working level because the axis of its wheel is above 2 m and are used for normal or emergency operations shall be equipped with wheels with chain.

- Valves whose wheel is below the operating level, will be provided with an extension stem that places the wheel to 500 mm on that level.

- When the P & ID indicates that the valves are in pipes that connect towers or tanks, they will be installed directly on these nozzles. In flap of towers or vertical vessels, the valve will be placed outside of the flap.

- The stem valve will never be rotated below the horizontal, being preferred the vertical position or closest to this.

- The battery limit valves to flare gas collectors are designed with the wheel horizontally or down.

- The valves, which in their operation are related to a next level indicator, will be located as close to it and at the same operating level, so that the instrument can be observed while the operator manipulates.

- Check valves are preferably installed horizontally or vertically with upstream, whenever the type of valve permits, but never downstream.

- All safety valves shall be accessible from the ground or platform, trying to use platforms that are necessary for other reasons.

- The battery limit valves, if any, will be grouped to be controlled from the ground or from platforms.

- Flare line battery limit valves should be installed horizontally with the stem in the downward position to avoid free falling of the gate and blockage of the flare system.

14.- INSTRUMENTS

- Displacement type level instruments and levels of glass will be positioned so that they are accessible from ground, platform or a permanent ladder. The levels of glass will be readable from the ground or platform, where is possible.

- Control valves are accessible from the ground or platforms. Generally they will be located at sight of the instruments or indicators that show control variables.

- Orifice flanges will be preferably placed horizontally. Orifice flanges may be installed vertically, when flow is upward in liquid services and downward for gases services and steam water. The intakes of the orifice flanges are oriented according to the standard SH-29 of **Intecsa-industrial**.

- Temperature indicators shall be visible from the ground, a platform or a ladder. All temperature connections must be accessible by a portable scale, and the highest by platforms and stairways.

- Local pressure indicators will be visible from the ground, fixed ladder or platform. All pressure instruments must be accessible by portable ladders, and the highest using fixed ladders and platforms.

15.- VENTS AND DRAINS

- In the P & ID only will display the vents and drains required for process operations. All will carry valves and will be of the size indicated by the Process Department.

- All high points of all lines must be vented and all low points should be drained except when the line can be vented or drained otherwise.

- Before each control valve it will be installed a drain to relieve the pressure prior to disassembly for maintenance.

- In general, all equipments will have ventilation and drainage, only if it can be done through a line will be allowed to be missing. The material of the vents and drains will agree with the specification most severe of all the pipes that connect to the equipment.

- To drain the steam pipes, steam traps will be used, which are placed along the collector at dead ends, and at the bottom of riser pipes. We recommend the use of steam traps with integrated valves.

- To protect from dirt the steam traps it will be placed steam traps, according to standard SH-3 of **Intecsa-Industrial** and strainers (the latter depending on the type of trap if required).

- The discharge of steam traps, may be to the condensate collector (preferably), or directly to the floor, as specified for the project.

16.- SERVICE SYSTEMS

16.1. Plant Air

- The plant air will generally be used in the following services

  a. As auxiliary source for instrument air.

  b. For processes when directed the P & ID.

  c. For utility stations.

  d. For pneumatic tools, hoists and equipment that are not shown in the diagrams.

- The plant air intakes from the main header will be made at the top of the main header and will be provided with a shut-off valve.

- In branches longer than 15 m, a block valve should be placed as close as possible to the header, even if another one is placed close to the equipment.

16.2. Instrument Air

- The instrument air collector will service all areas required by the Instrumentation Department. They have provided intakes with valve in the top of the collector, from where the Instrument Department will connect for service instrument air drum or directly to instruments.

16.3. Steam and Condensate

- All branch connections to main steam collectors located in horizontal will be as close as possible to the upper generatrix. These branches will have a block valve in the horizontal section.

- Pockets or low points, where condensate can focus, will be minimized; if this happens, traps will be installed on steam traps to remove the condensate. No steam trap will receive more than one condensate connection.

16.4. Steam tracing and jacketed

- When necessary to maintain the fluidity of the product or to prevent the effects of frost in piping and equipment, will be used steam tracing or jacketed steam, hot water or thermal conductor oils.

- To reduce heat tracing, it shall take into account the following recommendations:

  a. Locate susceptible frost pipes under cover or interior.

  b. Place underground water pipes without a continuous circulation.

  c. Minimize pipes with dead ends containing liquids easily solidification.

  d. Use bypass between supplies and return piping to maintain circulation flow.

- The designs of steam tracing or jacketing is made in according to the specifications and standards of **Intecsa-Industrial**.

16.5. Cooling water

- The pipes for cooling water will be distributed air or buried as appropriate. When are buried, the cutoff valves of the battery limit and instruments will be installed in drain boxes.

- All valves are above the floor together with consuming equipments.

- Blind flanges and drains will be placed in all end collectors.

- The branches are connected at the top or bottom of the collector, depending on the specific design requirements.

16.6. Fire water

- The fire service network will be designed and detailed by the Department of Mechanics: FFS. In projects as required by the customer, the Piping Department will capture this design in the 3D model or at the piping drawings if it was included in the scope of tender. Otherwise it will be subject to a Change Order.

16.7. Off steam

- Off steam is used to control a possible fire inside the furnaces, also be installed around the flanges of the head changer and fire-endangered if required.

- This steam will be permanently connected to the combustion chamber, head tube furnaces where these heads are removable, intakes of furnaces, combustion chamber and convection section where indicate in the P & ID.

- The control station or cutoff valve must be located at least 15 m from the furnace.

16.8. Fuel Gas

- Feed lines of burners will connect with the upper generatrix of the collector and the control station will be located near the furnace and preferably in the ground. In gas fuel burners it will be placed a connection for vaporized downstream the control blowing burner nozzles.

- Quick-closing valves of burners should be in the platform so that they can be operated while watching the fire through the peephole.

16.9. Fuel Oil

- The control station of fuel oil will be located near the furnace, reaching to burners individual branches. Quick-closing valves shall be located so that they can be accessed from the observation peepholes.

- Burner pipes are arranged to allow that burner nozzles fit or dismount without dismantling the pipe.

- All collectors will form loops (for round trip) near the furnace. This will have no dead points and will be traced with steam.

16.10. Fog Steam (Atomizing Steam) for Fuel Oil

- The atomizing steam is used when fuel is fuel oil, installing cutoff and check valves. Steam traps are placed with all steam traps stations in low points.

- The shots to each burner will be connected in the upper generatrix of the collector. It will mount a quick-closing valve for each burner and will be accessible from the observation peephole.

16.11. Utility Station and Eyewash Shower

- Utility stations shall be provided so that using 15 m hoses entire unit be covered. Preferably be placed next to the pillars of the rack or structures.

- For equipment located above 10 meters, utility stations will be installed higher if required the P & ID's. Service stations will be as specified in the standard SH-8 of **Intecsa-Industrial** and the necessity of the different services (air, steam, nitrogen, water) for each station will also be indicated in the P&IDs.

- In points where is needed, as areas of chemicals products, safety showers and eyewash showers are installed as SH-41 standard of **Intecsa-Industrial** or per ANSI / ISEA Z 358-1.

16.12. Samplers

- The connections for samplers should be made in the top of horizontal or vertical pipes to prevent contamination. In the case you need to cool the sample it will be installed a cooler as standard SH-28 of **Intecsa-Industrial**.

- The samplers are usually double valve, cutoff valve and control valve.

- When the sample is in a highly viscous product, is connected a steam line near the cutoff valve and the process pipe, to make a vaporization and clean the sample.

- All sampler requirements will be indicated in the P & ID's.

17.- THERMAL EXPANSION AND FLEXIBILITY. STRESS ANALYSIS

- Forecasts for thermal expansion and contraction will be in all lines in accordance with ASME B31.3, B31.1, B31.12 or B31.8.

- The basis of calculation for the expansion will be the maximum operating temperature and / or other more severe temporary conditions.

- The effects of the expansion will be resolved with arms, lyres and will be exceptionally resorted to expansion joints.

- The thrust forces and torques that pipes exert on mechanical equipment shall not exceed the values required by the equipment manufacturer.

- It will be taken into account in all cases, the stresses due to hydraulic test, washing of pipes with water and chemical cleaning, with pipe attached, indicating explicitly when passing information to other departments of these conditions, in addition to the operating.

18.- SUPPORTS, ANCHORAGES, GUIDES AND TOPS

- All pipes will be adequately supported by hanging, support brackets or other structural design.

- Pipes which need to be periodically dismantled will be supported so that elements can be extracted without installing temporary supports.

- Lines requiring isolation shall be provided with skids as indicated. The length of the skids will allow the longitudinal displacement of the pipe without the support runs out.

- All anchors are designed for normal and abnormal conditions taking the most restrictive of them. Normal conditions are defined in ASME B 31.3.

- Special attention will be paid to the support of lines subject to vibrations, supporting independently of foundations of the equipment that motivates them.

- The supported will consider hydraulic test conditions, water washing, chemical cleaning and steam cleaning.

19.- INSULATION AND PAINT

- The insulation is applied to pipes which require heat loss limitation or gain of process and economic effects, for personnel protection, cold maintenance or condensate removal. The thickness and type as the corresponding specification will be published by the Process Department and shall be included in the list of lines to display in drawings and piping material listings.

- The pipeline will be protected and painted in accordance with the appropriate specification published by the Piping Department and its type must be included in the list of lines to display in drawings and piping material listings.